M Tech (AIML)_Machine Learning _Assignment 1

Problem Statement 5

With the rapid expansion of digital payment infrastructures, financial systems are increasingly reliant on **behaviour-driven risk scoring engines** rather than rule-based fraud filters. Modern fraud detection systems must move beyond binary classification and instead infer **latent behavioural anomalies** under uncertainty.

You are given a transactional dataset capturing **customer financial behavior, contextual metadata, and merchant attributes**. Rather than treating fraud detection as a purely supervised learning problem,

your objective is to: **Design, analyse, and justify a robust machine learning pipeline that models transactional risk as a function of behavioural deviation and contextual signals.**

1. Exploratory Behavioral Analysis

Instead of basic EDA:

a. Perform **behavioral segmentation analysis**:

- Identify at least **3 distinct customer behavior clusters** (e.g., high-frequency low-value vs low-frequency high-value).

b. Analyze:

- How fraudulent vs non-fraudulent samples differ **in distribution space**, not just averages.

c. Construct at least **one composite behavioral indicator** (e.g., normalized spend velocity).

d. Critically evaluate:

Does fraud appear as a **separable class** or as **anomalous behavior embedded within normal patterns**?

2. Feature Engineering as Hypothesis Building

You are NOT allowed to blindly apply transformations.

a. Formulate **at least 3 hypotheses**, e.g.:

- "High velocity + low merchant reputation increases fraud likelihood"

b. Design features to test them:

- Interaction terms
- Temporal encoding
- Behavioral ratios

c. Compare:

- Raw vs engineered feature performance using **feature importance methods** (at least 2 methods)

d. Justify:

Which features are **causal proxies vs spurious correlations**?

3. Data Conditioning and Robustness

a. Instead of basic cleaning:

- Analyze **distribution skewness and tail behavior**
- Apply transformations **only if justified statistically**

b. Detect and treat:

- Outliers using **model-aware methods** (not just IQR)

c. Discuss: How preprocessing choices affect **model bias and variance**

4. Model Design Under Constraints

You must treat this as a **risk modeling system**, not just classification.

a. Train-Test Strategy

- Compare:
  - Random split
  - Stratified split
  - Time-aware split (if applicable)

Explain implications for **real-world deployment**

b. Model Development

You must implement:

1. A **linear model** (Logistic Regression or equivalent)
2. A **tree-based model** (Decision Tree / Random Forest)
3. One **advanced model** (choose one):
   - Gradient Boosting
   - XGBoost / LightGBM

    - OR Isolation Forest (if framing as anomaly detection)

c. Hyperparameter Optimization

Use **cross-validation**

Compare: Grid Search vs Random Search

Explain:Why certain hyperparameters matter in **risk-sensitive systems**

5. Evaluation Beyond Accuracy

a. Evaluate using:

- Precision, Recall, F1-score
- ROC-AUC
- Confusion Matrix

b. Additionally:

- Analyze **false positives vs false negatives trade-off**
- Define: Cost of missing fraud vs falsely flagging users

c. Provide a **cost-sensitive evaluation metric**

6. Model Interpretability & Trust

a. Use:

- Feature importance (global)
- SHAP / LIME (local explanation)

b. Answer:

Can your model be trusted in a regulatory environment?